

Luiss

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degli Studi Sociali Guido Carli

Multiprocessing

Introduction to Computer Programming

Management and Artificial Intelligence

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How does a computer execute code?

The code of a program is executed by a CPU. A CPU has **one or more cores**, each of which can execute a single instruction at a time.

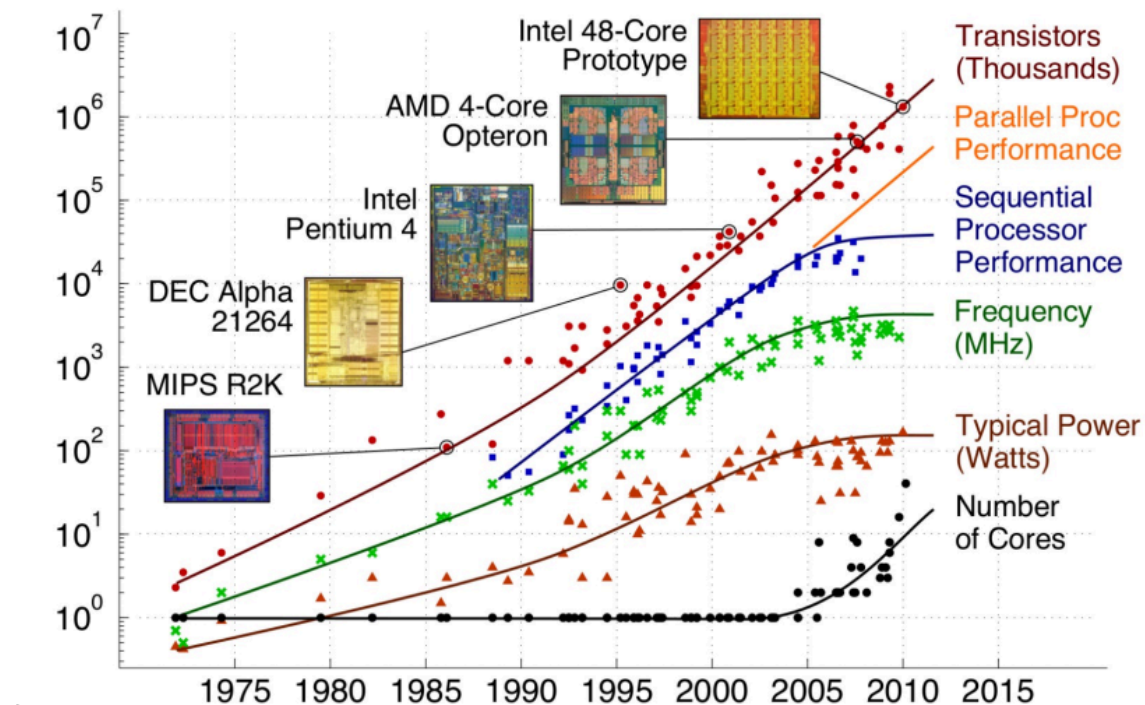
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How can make our code run faster?

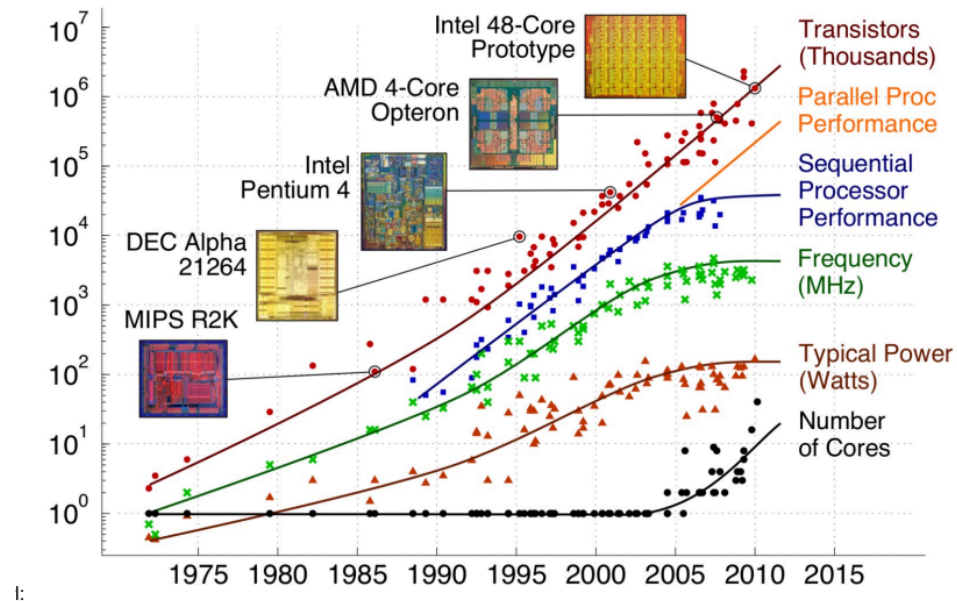
More transistors, more speed (20 years ago)

Moore's Law (1965) states that the number of transistors on a microchip doubles about every two years with a minimal cost increase. For a long time, **more transistors meant more speed**.



l:

Free lunch is over: more transistors, more cores



In the past, we could rely on the fact that the CPU speed would double every two years. This is no longer the case. The CPU speed has not increased significantly in the last decade. Instead, the **number of cores has increased**. The problem is that **most of the software is not designed to take advantage of multiple cores**.

Single Task Execution [single core]

If we run our Python program, the CPU will execute the code sequentially, i.e., one instruction at a time. This is known as *single task execution*:



The line in yellow is the execution of our Python program.

Multi Task Execution [single core]

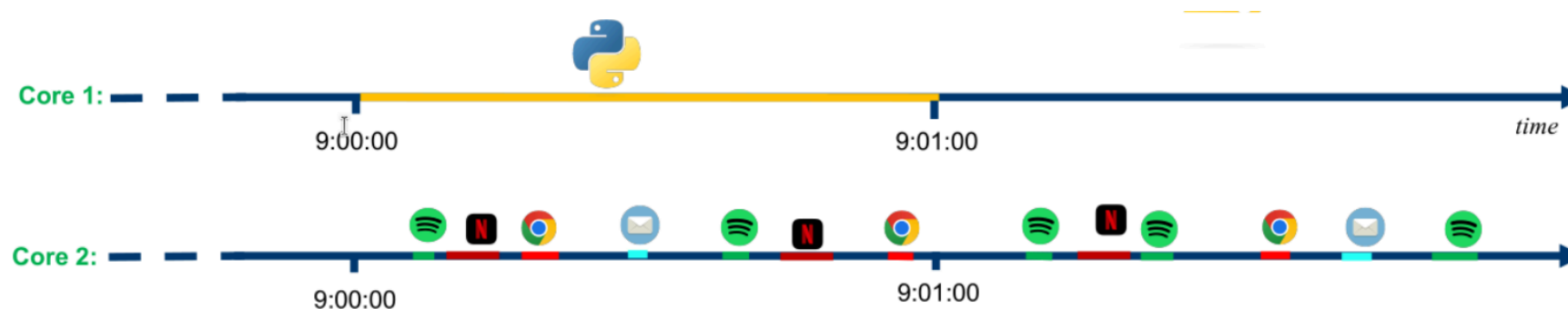
Modern operating systems allow for *multi task execution*. This means that the CPU can execute multiple tasks at the same time. This is achieved by rapidly switching between tasks. This is known as *time sharing*:



The line in yellow is the execution of our Python program. It is not a continuous line because the CPU is executing other tasks.

Multi Task Execution [multi core]

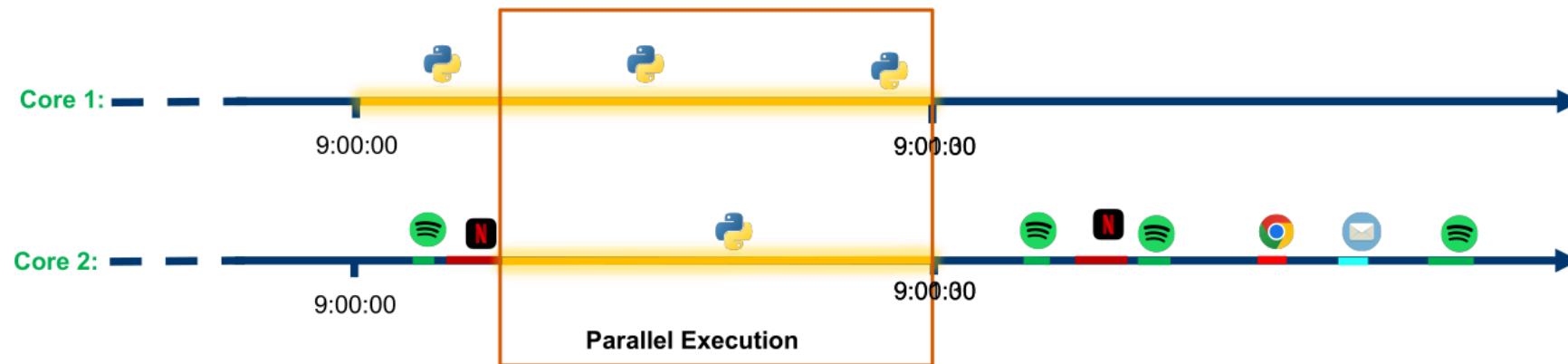
When we have multiple cores, we can execute multiple tasks at the same time. This is known as *multi core execution*:



The line in yellow is the execution of our Python program. It could be not a continuous line when the CPU is executing other tasks on the same core.

Parallelism

Parallelism is the ability to execute multiple tasks at the same time:



Processes

- A **process** is an instance of a program in execution
- Each process has its own:
 - **memory space** (variables, heap, stack)
 - **file descriptors**, environment, etc.
- The operating system schedules processes on available **CPU cores**

Relation to parallelism & concurrency:

- On **one core**, multiple processes are *concurrent* (time-shared)
- On **multiple cores**, different processes can truly run in **parallel**

In Python:

- The multiprocessing module creates **child processes**
- Each process has its own Python interpreter and its own GIL (Global Interpreter Lock)

Threads

- A **thread** is a lighter unit of execution **inside a process**
- Threads in the same process share:
 - the same **address space** (global variables, objects)
 - open files, sockets, etc.

Relation to concurrency & parallelism:

- Threads are often used to express **concurrency**:
 - many activities *in progress* within the **same process**
- On multi-core CPUs, different threads *could* run in **parallel**

Threads vs. Processes

Characteristics	Threads	Processes
Creation	Lightweight	Heavyweight
Communication	Shared memory	IPC
Overhead	Low	High
Parallelism	Within a process	Inter-process
Scalability	Good	Better
Code complexity	Simpler	More complex
System resources	Shared	Independent
Data race issues	Possible	Less likely

A Python program likely uses a single core

Most Python code is designed to run on a single core. Up to now, we have we have written code that runs on a single core. Besides technical reasons, the main problem is that writing code that works on different cores is complex and error-prone. For instance:

1. How do we split a larger task into multiple subtasks?
2. How do we synchronize the subtasks?
3. How do we collect the results?

These problems are not easy to solve. After 50 years, there is still no general solution to these problems. In other words, we cannot take a traditional program designed to run over a single core and make it run over multiple cores without rewriting it.

Parallelism in Python

There are several ways to write parallel code in Python. The most common ways are:

1. **threading**: the threading module allows us to run multiple ***threads*** within a single process. Different **threads can easily share data because they share their memory space**.
2. **multiprocessing**: the multiprocessing module allows us to run multiple processes. **Each process has its own memory space** and runs in its own memory space. Hence, **sharing data is expensive**.

Threads in Python: the GIL Problem

In CPython, i.e., the reference implementation of Python:

- The **Global Interpreter Lock (GIL)** is a mutex that protects access to Python objects, preventing multiple threads from executing Python bytecode simultaneously

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 - For **CPU-bound tasks**: threads provide **no performance benefit** and may even add overhead due to context switching
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Workaround: Use multiprocessing for true parallelism in CPU-intensive workloads

Multiprocessing is not threading

- We use the multiprocessing module to create **separate processes**
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 - No GIL issues
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Major challenge: Sharing Data Between Processes

- Different processes **do not share memory** by default
- We need special mechanisms to share data across processes:
 - Shared memory objects
 - Queues and Pipes
 - Manager objects

A small CPU-bound task

Let us consider a syntetic CPU-intensive task: executing 2 times `compute_something(2500000)`

```
import time

def compute_something(n):
    r = 0
    # this is just a synthetic computation
    for i in range(n):
        r += (200 ** 200) % (1000 + i)
    return r # not a very useful computation

start = time.time() # get the current time
result = compute_something(2500000) # first computation
result = compute_something(2500000) # second computation
end = time.time() # get the current time

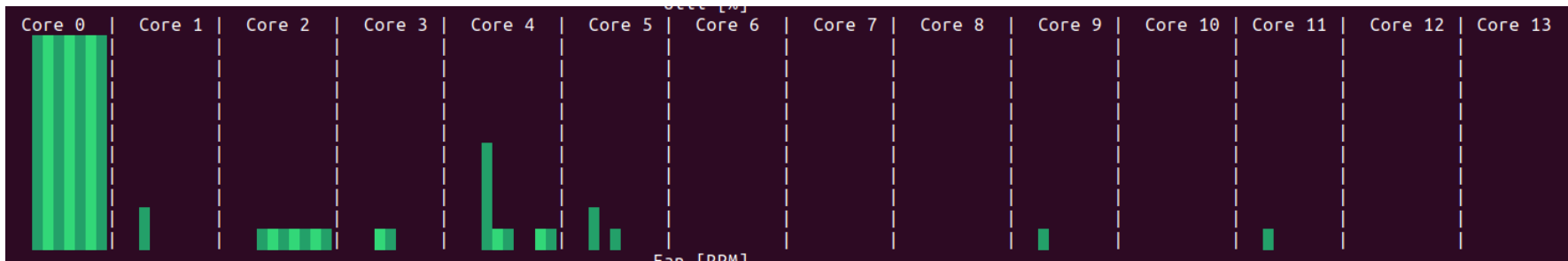
print("Elapsed time:", end - start, "seconds")
```

Elapsed time: 4.413878440856934 seconds

CPU-bound tasks: sequential execution

We are executing `compute_something` twice sequentially. This is a CPU-bound task.

Only one core is used:



threading: parallel function execution?

```
import threading
from worker import compute_something
import time

t1 = threading.Thread(target=compute_something, args=(2500000,))
t2 = threading.Thread(target=compute_something, args=(2500000,))

start = time.time()
t1.start()
t2.start()
# wait for the process to finish
t1.join()
t2.join()
print("Elapsed time:", time.time() - start, "seconds")
```

Elapsed time: 4.694298982620239 seconds

BUT WAIT... **IT IS NOT FASTER!**

WHY?

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WHY?

As we already said, the GIL prevents parallel execution of Python code.

multiprocessing: parallel function execution

```
import multiprocessing
import time
from worker import compute_something

p1 = multiprocessing.Process(target=compute_something, args=(2500000,))
p2 = multiprocessing.Process(target=compute_something, args=(2500000,))

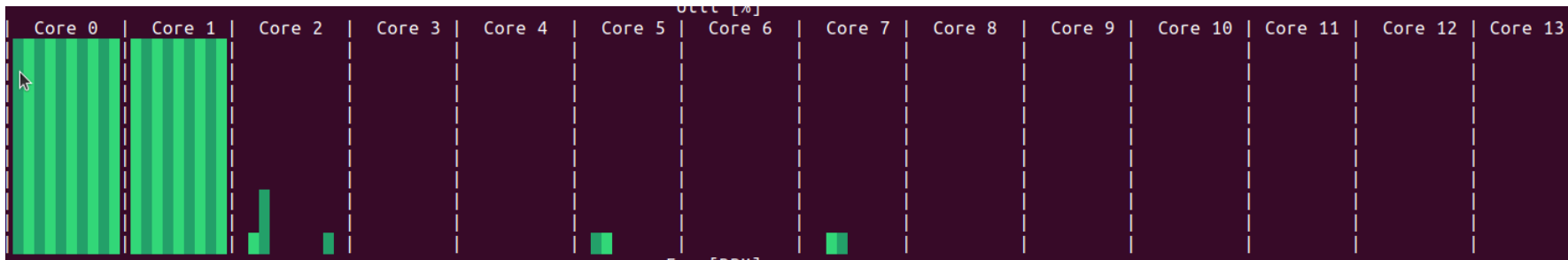
start = time.time()
p1.start()
p2.start()
# wait for the process to finish
p1.join()
p2.join()
print("Elapsed time:", time.time() - start, "seconds")
```

Elapsed time: 2.499671459197998 seconds

CPU-bound task: parallel execution

We are executing one `compute_something` on each core.

Two cores are used:



multiprocessing: work assignment

What if we have more tasks than processes?

multiprocessing: work assignment

What if we have more tasks than processes?

`multiprocessing.Pool` allows us to create a pool of processes. We can then assign tasks to the pool. The pool will take care of assigning the tasks to the processes.

For instance:

```
import multiprocessing
from worker import compute_something

pool = multiprocessing.Pool(2)
inputs = [2500000, 2500000, 2500000] # we want to compute the same thing three times
results = pool.map(compute_something, inputs)
print(results)
```

```
[1561361386212, 1561361386212, 1561361386212]
```

multiprocessing: issues about sharing data

The processes do not share memory. By default, multiprocessing takes care of:

1. Transferring the data related to arguments (in our example: `n=2500000`)
2. Transferring the data related to the return value (in our example: `r`)

These operations can be expensive when the data to move around is large.

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These operations can be expensive when the data to move around is large.

Depending on what we are doing, we may find a way to avoid such a data transfer. For instance, when our function needs to process a file and compute the results, instead of passing the data of the file and get back the data of the processed file, we can just read and write the file from the disk and pass to the function only the file name.

There are several cases when we would like to share data between processes. For instance, we may want to share a large dataset between processes. In such cases, we can use the `multiprocessing.Manager` class to build data structures (such as lists) that are shared.

However, sharing data is expensive and should be avoided when possible.

Suppose we want to use a list between processes:

```
def compute_something_with_index(index, l, n):  
    r = 0  
    for i in range(n):  
        r += (200 ** 200) % (1000 + i)  
    l[index] = r # we store the result in the list
```

```
import multiprocessing  
from worker import compute_something_with_index  
  
l = [None, None] # list  
p1 = multiprocessing.Process(target=compute_something_with_index, args=(0, l, 2500000))  
p2 = multiprocessing.Process(target=compute_something_with_index, args=(1, l, 2500000))  
  
p1.start() # start the process  
p2.start() # start the process  
p1.join() # wait for the process to finish  
p2.join() # wait for the process to finish  
print(l)
```

[None, None]

The list `l` has not changed! This is because the list `l` is not shared between the processes. Instead, each process has its own copy of the list `l`.

We can use the multiprocessing.Manager class to create a shared list:

```
import multiprocessing
from worker import compute_something_with_index

with multiprocessing.Manager() as manager:
    l = manager.list() # create a shared list
    l.append(None)
    l.append(None)
    p1 = multiprocessing.Process(target=compute_something_with_index, args=(0, l, 256))
    p2 = multiprocessing.Process(target=compute_something_with_index, args=(1, l, 256))
    p1.start()
    p2.start()
    p1.join()
    p2.join()
    print(l)
```

[1561361386212, 1561361386212]

We can use the `multiprocessing.Manager` class offers several data structures that can be shared between processes:

- `list`
- `dict`
- `Value`
- `Array`
- ...

They work like the normal data structures but they are shared between processes. However, sharing data is expensive and should be avoided when possible.

Parallelism versus Artificial Intelligence

Should you use parallelism in your data science projects?

Most of the time, unlikely. At least, not directly. What it may happen is that you may use a library that uses parallelism under the hood. For instance, `parallel_pandas` uses parallelism when you apply a function to a `DataFrame`.

Hence, you should have an idea of what we mean by parallelism and how it works.

Summary

- Modern CPUs have **multiple cores** → potential for **parallel** execution
- **Concurrency** vs **parallelism**:
 - concurrency = many tasks in progress (may be interleaved)
 - parallelism = tasks running at the same time on different cores
- **Threads** (threading):
 - good for **I/O-bound** tasks
 - limited for **CPU-bound** tasks in CPython due to the **GIL**
- **Multiprocessing** (multiprocessing):
 - separate processes can use multiple cores
 - use `Process`, `Pool`, and data-sharing tools (`Value`, `Array`, `Manager`, `Queue`) to structure parallel programs.

